

CCSW-225

Human-Computer Interaction

Design Project

Section SE3

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1. The general Project idea and Title

The Fresh Bite App is a smart cooking and nutrition assistant that helps users prepare healthy meals based on the ingredients they have at home. It provides personalized recipe suggestions, automatic calorie and nutrition calculation, and ingredient substitutions for dietary restrictions and allergies.

Users can plan meals, generate shopping lists, and sync data with fitness apps to track their health goals. Whether you're a health-conscious individual, fitness enthusiast, or beginner cook, this app makes healthy eating easy, efficient, and accessible.

2. Who are the targeted users?

- a. Healthy Lifestyle members
- b. Beginner cooks
- c. People with dietary restrictions
- d. People with allergies
- e. Home cooks
- f. Athletes
- g. People who want to explore new recipes

3. What are the main problems we are trying to solve?

- a. Difficulty completing recipes due to missing ingredients.
- b. Lack of knowledge about ingredient substitutions and lack of nutrition awareness.
- c. Challenges finding safe substitutes for allergy sufferers.
- d. Need for nutritional guidance for beginner athletes.
- e. Difficulty in Finding Healthy Meal Ideas.

4. How are you going to collect the data from the users?

We will be collecting data from users by using two techniques which are Interviews and questioners.

5. How are you going to find users?

We're going to find users by spreading our questioner online to target individuals interested in healthy eating, cooking, and fitness

To really understand what people need, we're going to connect with a few different groups. We'll be talking to gym members, people who are focused on their diets, and individuals who are active in online fitness communities, basically, anyone who's already tracking what they eat. We also plan to make interviews with people we know – friends, family, classmates – to get a sense of the everyday challenges they face when it comes to planning their meals.

6. What interview questions are you going to ask?

- What are the biggest challenges you face when trying to prepare healthy meals?
- What features do you consider essential in an app that helps you prepare healthy meals?
- If you are allergic to an ingredient, what challenges do you face?
- Have you ever used any apps to help with cooking or meal planning? If yes, which ones?
- How important is it for you to track calories and nutrition in your meals?
- Do you track your fitness or health goals using any apps? If yes, which ones?
- How do you currently find ingredient substitutes when needed?

7. The link to the online questioner

<https://forms.office.com/r/dewwLieJ4P?origin=lprLink>

8. The raw data collected from the users

The Excel link

https://jeddahu-my.sharepoint.com/:x:/g/personal/2317129_uj_edu_sa/EeFdEfiLYyhEgpPvUsk1zwYB78xOP3DY4TxBRZxaNHrpDA?e=ScIzoc

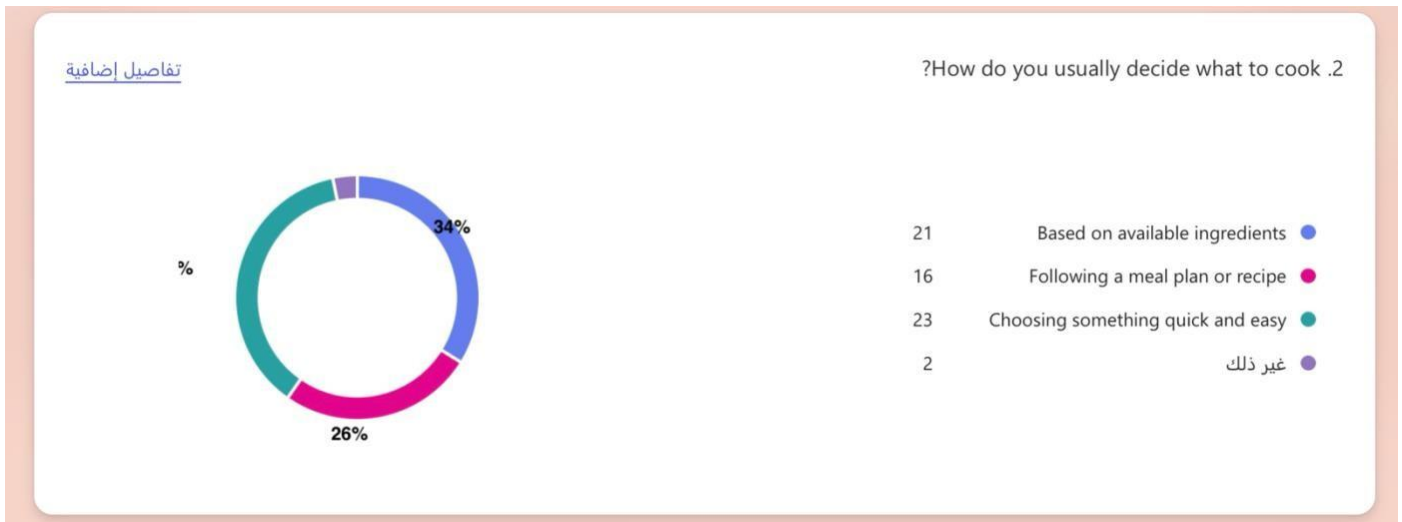
The Interview Analysis

Interviewer	Interviewee	Analyst comment
Abrar	Name: Salma Alharbi Age: 32 Job: hospital manager secretary Another information a mother of four children	Salma Alharbi, a mother of four and a hospital manager secretary, struggles to get her kids to eat healthy. “They don’t like vegetables unless they’re fried,” she says. She wishes for an app with kid- friendly recipes and healthy alternatives to fast food. Instead of meal-planning apps, she finds ideas on TikTok. Salma focuses more on sugar than calories, knowing its impact on children’s health. She stays active by walking and tracking her steps. A smart cooking app with fun, healthy recipes and easy ingredient swaps would make her cooks much easier.
Reema	Name: Raghad Al Zharani Age: 25 Job: Hr She follows a healthy and active lifestyle	As someone who leads a healthy lifestyle, Raghad feels that one of the main obstacles she encounters when cooking healthful meals is the availability of healthy ingredients, as it can occasionally be challenging to locate appropriate substitutes in the marketplace. Time is another issue because making nutritious, balanced meals takes more preparation and planning than ordinary meals. According to her, a healthy meal preparation app should have quick and simple recipes with comprehensive nutritional data for every meal, along with a customization feature that lets her modify meals according to the items she has on hand. Raghad struggles to find appropriate substitutes because she is lactose intolerant, particularly when experimenting with new dishes. To prevent any adverse reactions, she thus needs trustworthy recommendations for substitute substances. She has utilized applications like MyFitnessPal to measure her caloric intake and Mealttime to plan her weekly meals in an efficient manner. For her, keeping track of calories and nutrients is crucial since it helps her reach her health objectives, like gaining muscle or losing weight, while making sure her meals are balanced. Raghad also uses applications like MyFitnessPal and Google Fit to track her daily activities. She tracks her gym workouts, which include cycling and running, using Strava. She typically looks up reliable suggestions online or in specialist nutrition communities when she wants ingredient substitutes.

Lamis	Name: Raghad Baafif Age: 20 Student Individual following a diet and wanting to track calories and nutritional information precisely	One of the biggest challenges I face in preparing healthy meals is planning to ensure variety without repetition while making sure that all the necessary ingredients are always available. Time constraints also make quick meal preparation difficult. I prefer apps that offer features such as suggesting recipes based on available ingredients, providing accurate nutritional information, allowing meal customization according to fitness goals, suggesting substitutes for unavailable ingredients, and having a user-friendly interface with a barcode scanner for easy input. I also struggle with reading food labels when looking for safe ingredients due to allergies. I have tried apps like MyFitnessPal and Miran, but they lack accuracy and do not cover all common foods. Tracking calories and nutritional values is very important to me to ensure a balanced intake of protein, carbohydrates, and fats in line with my goals, so I use apps like MyFitnessPal and Health to monitor my nutrition and daily activity. When I need ingredient substitutes, I rely on personal experience or online research.
Jana	Name: Leen Alghamdi Age: 21 Medical Student Another information Gym athlete and healthy lifestyle living	Leen Alghamdi, a 21-year-old athlete and medical professional, faces challenges in maintaining variety in her meals while staying within her calorie deficit. She often struggles with finding new recipes that align with her fitness goals, leading to repetitive meal choices. When considering a meal-planning app, she prioritizes features that offer personalized recipes based on her calorie needs and available ingredients. Additionally, she values detailed nutritional information and ingredient swap suggestions. These insights highlight the need for a flexible and user-friendly meal-planning solution tailored to individuals focused on fitness and nutrition.

Enas	Name: Lmeaa Alsubhi Age: 24 Master Student She's a beginner cook, new to the healthy lifestyle, and seeking to prepare fast, nutritious meals due to her busy academic schedule.	Lmeaa, a 24-year-old master's student, is new to the healthy lifestyle and aims to start cooking fast, nutritious meals this year due to her demanding academic schedule, having previously relied on fast food and outside meals. As a beginner cook, she finds it challenging to choose the right ingredients for healthy meals and often struggles with finding suitable substitutes when items are unavailable. She values an app that offers personalized meal suggestions based on the ingredients she has, along with clear, step-by-step instructions and detailed nutritional information. Tracking calories and nutrition is essential to her as she navigates this new healthy lifestyle, using MyFitnessPal to monitor her progress. A key problem the Fresh Bite app could solve for Lmeaa is saving her time and reducing stress by providing quick, easy-to-follow recipes and ingredient substitutions, making it easier for busy beginners like her to adopt and maintain a healthy eating routine.
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9. A simple visualization of the data collected (graphs, charts, ...)

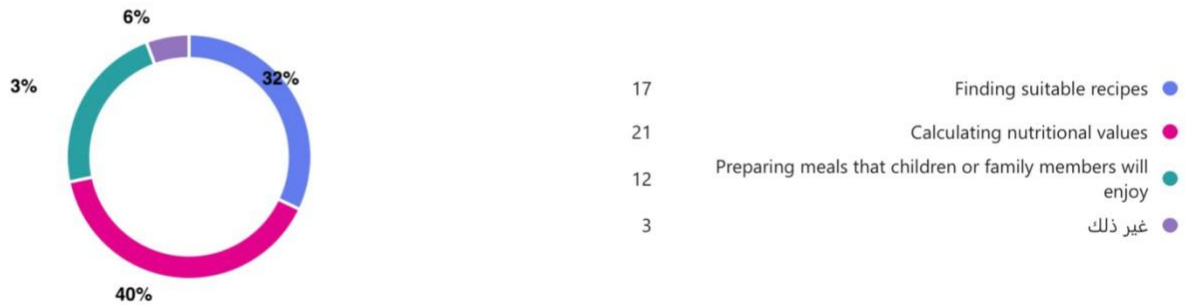


How do you usually decide what to cook?

- **Choosing something quick and easy:** 34% (23 votes)
- **Based on available ingredients:** 31% (21 votes)
- **Following a meal plan or recipe:** 26% (16 votes)
- **Other:** 3% (2 votes)

summary

- The most common approach (34%) is selecting something quick and easy.
- 31% decide based on available ingredients.
- 26% follow a structured meal plan or recipe.
- Only 3% use other methods.
- This indicates that convenience and availability are the biggest factors in meal selection, suggesting that an app focusing on quick meal ideas or ingredient-based suggestions would be useful.

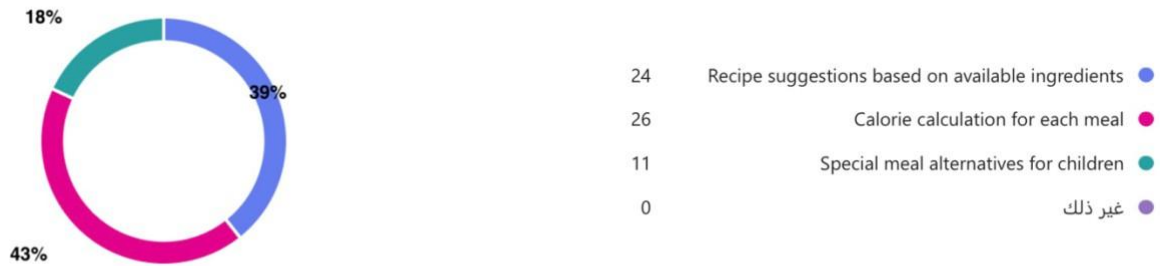


What is the biggest challenge you face when preparing healthy meals?

- **Calculating nutritional values:** 40% (21 votes)
- **Finding suitable recipes:** 32% (17 votes)
- **Preparing meals that children or family members will enjoy:** 23% (12 votes)
- **Other challenges:** 6% (3 votes)

summary

- The biggest challenge (40%) is calculating nutritional values, highlighting the need for an easy way to track meal nutrition.
- 32% struggle with finding suitable recipes, indicating demand for recipe recommendations.
- 23% face difficulty making meals enjoyable for children or family members, suggesting the importance of family-friendly meal options.
- A small percentage (6%) have other challenges.
- This suggests that an app focusing on nutrition tracking, meal recommendations, and family-friendly recipes would be highly beneficial.



Which feature would be most useful in a Healthy Food App?

- **Calorie calculation for each meal:** 43% (26 votes)
- **Recipe suggestions based on available ingredients:** 39% (24 votes)
- **Special meal alternatives for children:** 18% (11 votes)
- **Other:** 0% (0 votes)

summary

- Calorie calculation is the most requested feature (43%), indicating a strong focus on nutritional awareness.
- 39% prefer recipe suggestions based on available ingredients, showing demand for meal planning assistance.
- 18% prioritize special meal options for children, which is a smaller but still relevant audience.
- No one suggested other features, confirming that these three options align with user needs. This suggests that a nutrition-focused app with meal planning and calorie tracking would be highly valuable.

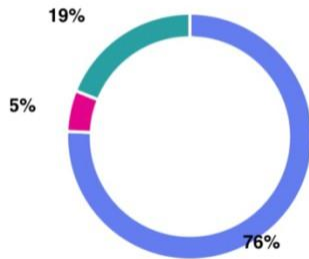


How often do you check the calorie content of your meals?

- **Daily:** 27% (10 votes)
- **Weekly:** 27% (10 votes)
- **Monthly:** 22% (8 votes)
- **Never:** 24% (9 votes)

summary

- The responses are evenly distributed, indicating varied interest in calorie tracking.
- 54% (daily + weekly) track calories regularly, suggesting a significant group is conscious about their intake.
- 22% check monthly, showing some level of interest but less frequent monitoring.
- 24% never check calories, meaning nearly a quarter of respondents do not prioritize calorie tracking. This indicates that while many value calorie tracking, a simplified or optional approach in an app would cater to all users.



28	Yes	●
2	No	●
7	Maybe	●

Would you find it helpful if the app suggested healthier alternatives for your ingredients?

- **Yes:** 76% (28 votes)
- **Maybe:** 19% (7 votes)
- **No:** 5% (2 votes)

summary

- A large majority (76%) support this feature, indicating a strong demand for ingredient substitution suggestions.
- 19% are open to the idea but not fully committed.
- Only 5% are not interested, suggesting that opposition to this feature is minimal.
- This indicates that adding a healthier ingredient suggestion feature would be widely beneficial and well-received by users.



How important is it for the app to provide special meal options for children?

- **Very important:** 38% (14 votes)
- **Somewhat important:** 43% (16 votes)
- **Not so important:** 19% (7 votes)

summary

- A majority (81%) consider this feature to be at least somewhat important.
- 38% think it's very important, while 19% see it as not so important.
- This suggests that including special meal options for children would be valuable to most users.



- 18 A daily meal plan based on my available ingredients
- 16 A flexible weekly meal plan with healthy options
- 3 No meal planning, I prefer to choose recipes myself

What type of meal planning assistance would you prefer?

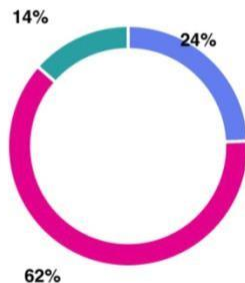
- **A daily meal plan based on my available ingredients: 49% (18 votes)**
- **A flexible weekly meal plan with healthy options: 43% (16 votes)**
- **No meal planning, I prefer to choose recipes myself: 8% (3 votes)**

summary

- Nearly half (49%) prefer a daily meal plan that considers available ingredients.
- 43% would rather have a weekly flexible plan with healthy options.
- Only a small percentage (8%) prefer no meal planning and want to choose recipes independently.
- This indicates a strong demand for structured meal planning, with a preference for ingredient-based and flexible plans.

تفاصيل إضافية

Would you be willing to pay for premium features (e.g., advanced meal plans, exclusive recipes, AI-based nutrition coaching)?



9 Yes ●
23 Maybe ●
5 No ●

Would you be willing to pay for premium features (e.g., advanced meal plans, exclusive recipes, AI- based nutrition coaching)?

- **Yes:** 24% (9 votes)
- **Maybe:** 62% (23 votes)
- **No:** 14% (5 votes)

summary

- A majority (62%) are open to paying but are undecided, meaning the value proposition needs to be clear.
- 24% are willing to pay, indicating a segment that sees direct value in premium features.
- Only 14% are not interested in paying, suggesting most users see potential benefits.
- This suggests that offering a freemium model with compelling premium features could be a successful approach.

10. List of 5 needs

11. List of 5 insights

12. Based on the needs and insights, define 5 tasks that your application will perform to meet the user needs

User	Need	Insights	Task
Home Cook (Salma)	Needs an app that helps her create kid-friendly, healthy meals with appealing presentations and smart ingredient substitutions.	She struggles to get her kids to eat vegetables and prefers recipes that make healthy food look fun and exciting. She also relies on TikTok for meal ideas, indicating that visual content is important for her.	Create a special page for children in the application that contains healthy foods presented in cartoon forms, as well as ways to replace unhealthy ingredients with other ingredients. Parents' suggestions for sharing their opinions
Individual with Allergy (Raghad)	Finding good swaps for ingredients: When Raghad tries new recipes, she needs a good place to find lactose-free options and other healthy stuff.	How lactose allergy changes what she eats: Raghad's need to avoid lactose really changes how she plans her meals and picks her ingredients, showing how key, it is to get the right swap tips.	Tool to find ingredient swaps: This tool gives users solid lactose-free choices and other healthy picks, based on what they need to eat
Gym Athlete (Leen)	She needs a meal-planning tool that offers personalized recipes based on her calorie goals and available ingredients, ensuring variety and flexibility.	Leen, an athlete and medical professional, finds it challenging to maintain meal variety while staying within her calorie deficit, often repeating meals due to a lack of new recipe ideas.	Create an app feature that suggests customized meal plans tailored to calorie intake and available ingredients, including alternative ingredient options to keep meals diverse and enjoyable.

Individual following a diet and wanting to track calories and nutritional information precisely (Raghad)	She needs an app that provides accurate calorie counts, displays detailed nutritional information for each meal, and helps customize meals according to her daily nutritional needs.	She struggles with maintaining meal variety while accurately tracking calories. She needs clear and precise nutritional information to ensure her meals are aligned with her diet goals.	Develop an app that helps Raghad track calories and nutritional information accurately, enabling her to follow her diet easily.
A beginner cooks, new to the healthy lifestyle, and seeking to prepare fast, nutritious meals due to her busy academic schedule (Lmeaa)	Lmeaa needs a simple way to explore and learn healthy cooking techniques while managing her limited time as a busy student. She needs assistance with step-by-step cooking guidance, time-saving meal suggestions, and meal planning to help her adopt and maintain a healthy lifestyle.	As someone new to cooking and the healthy lifestyle, Lmeaa often feels lost when trying to prepare meals efficiently. Her hectic schedule makes it difficult to plan meals ahead or spend long hours in the kitchen. This highlights an opportunity for an app feature that not only offers quick recipes but also teaches basic cooking techniques and helps users like Lmeaa build confidence in the kitchen over time.	Introduce a “Cooking 101” section in the Fresh Bite app that provides beginner- friendly tutorials, time- saving meal prep tips, and easy meal planning tools, ensuring that new cooks like Lmeaa can develop their skills while maintaining a healthy lifestyle without stress.

13. The tasks

Task 1:

Introduce a “Cooking 101” section in the Fresh Bite app that provides beginner-friendly tutorials, time- saving meal prep tips, and easy meal planning tools

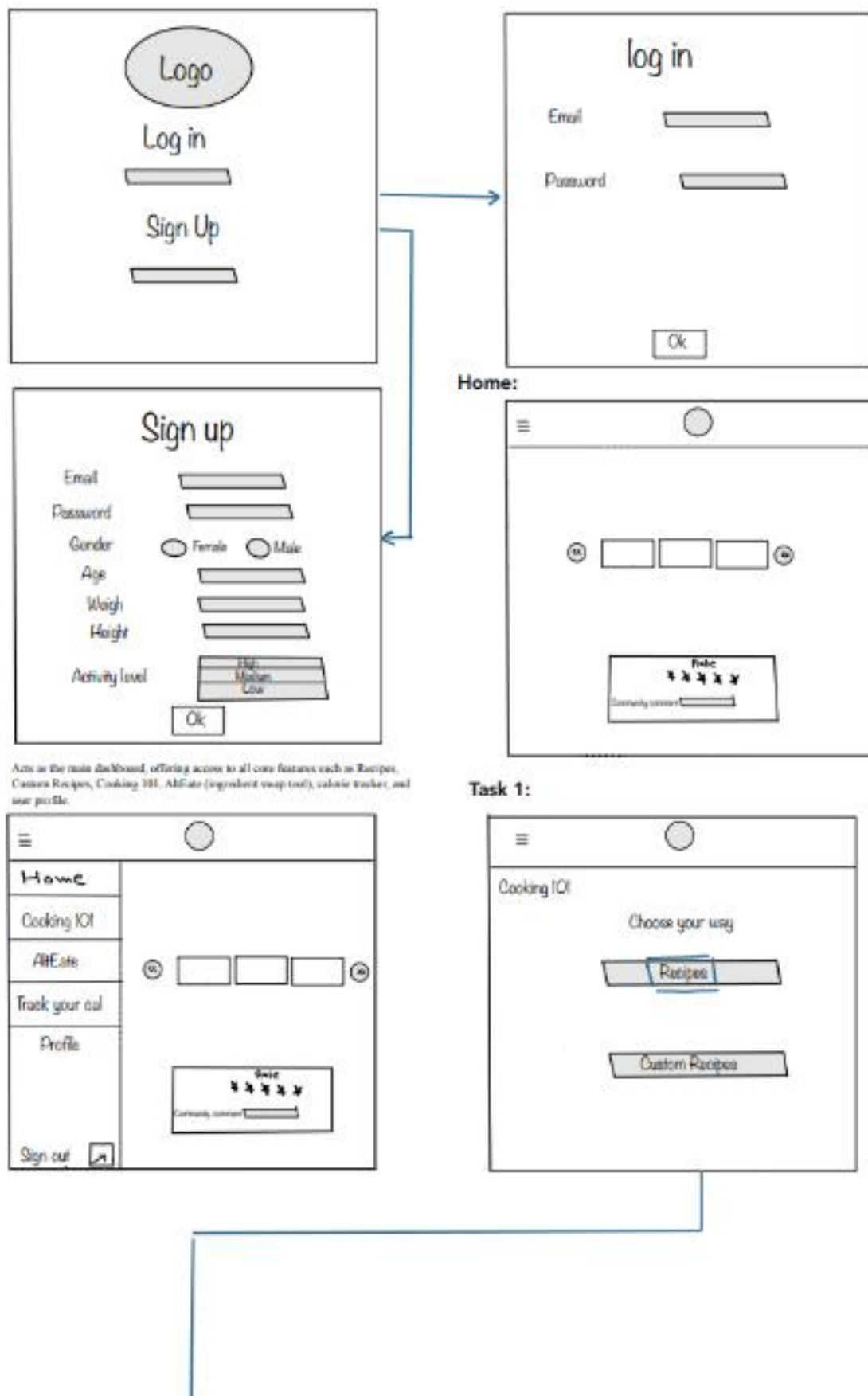
Task 2:

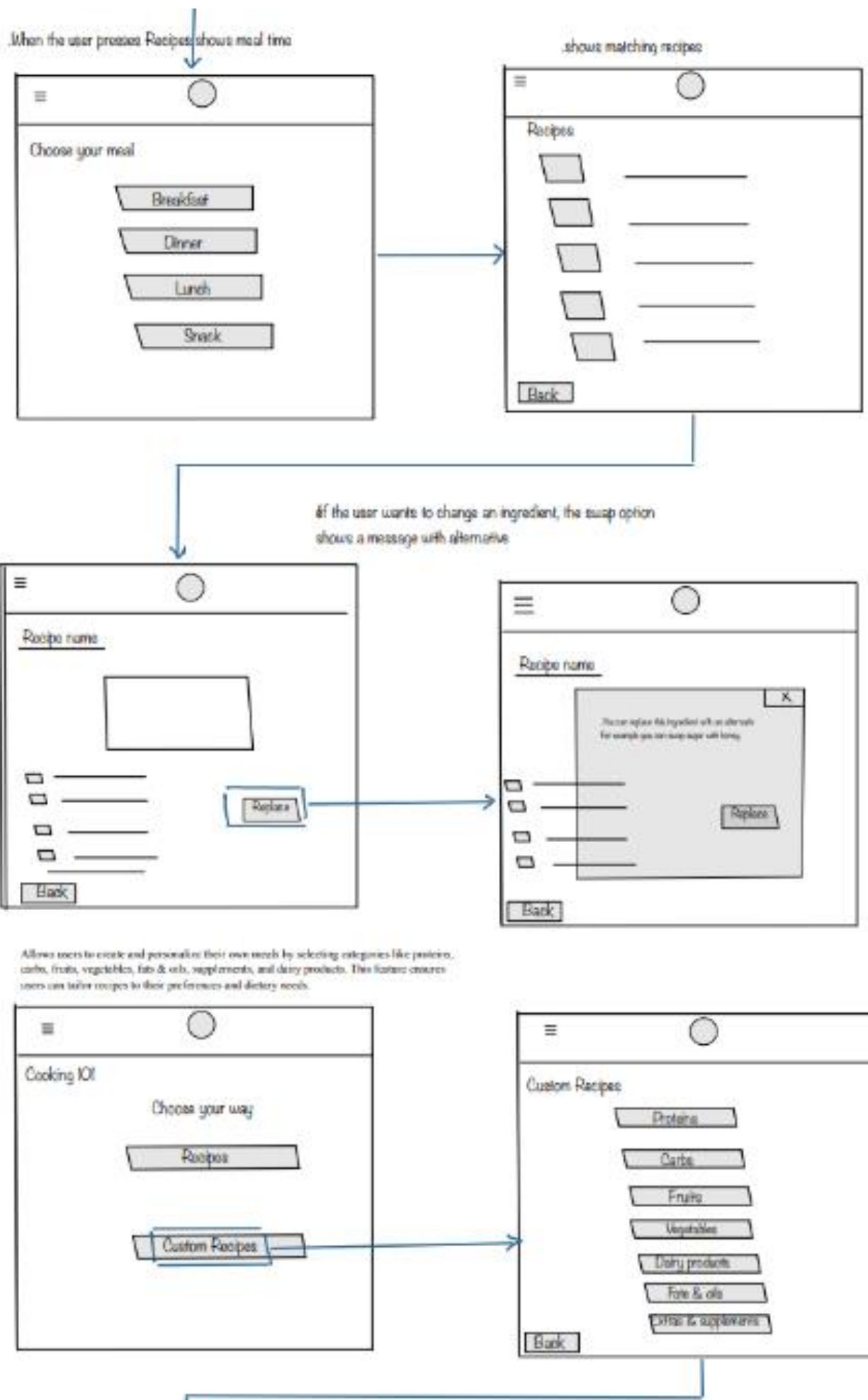
Tool to find ingredient swaps: This tool gives users solid lactose-free choices and other healthy picks, based on what they need to eat

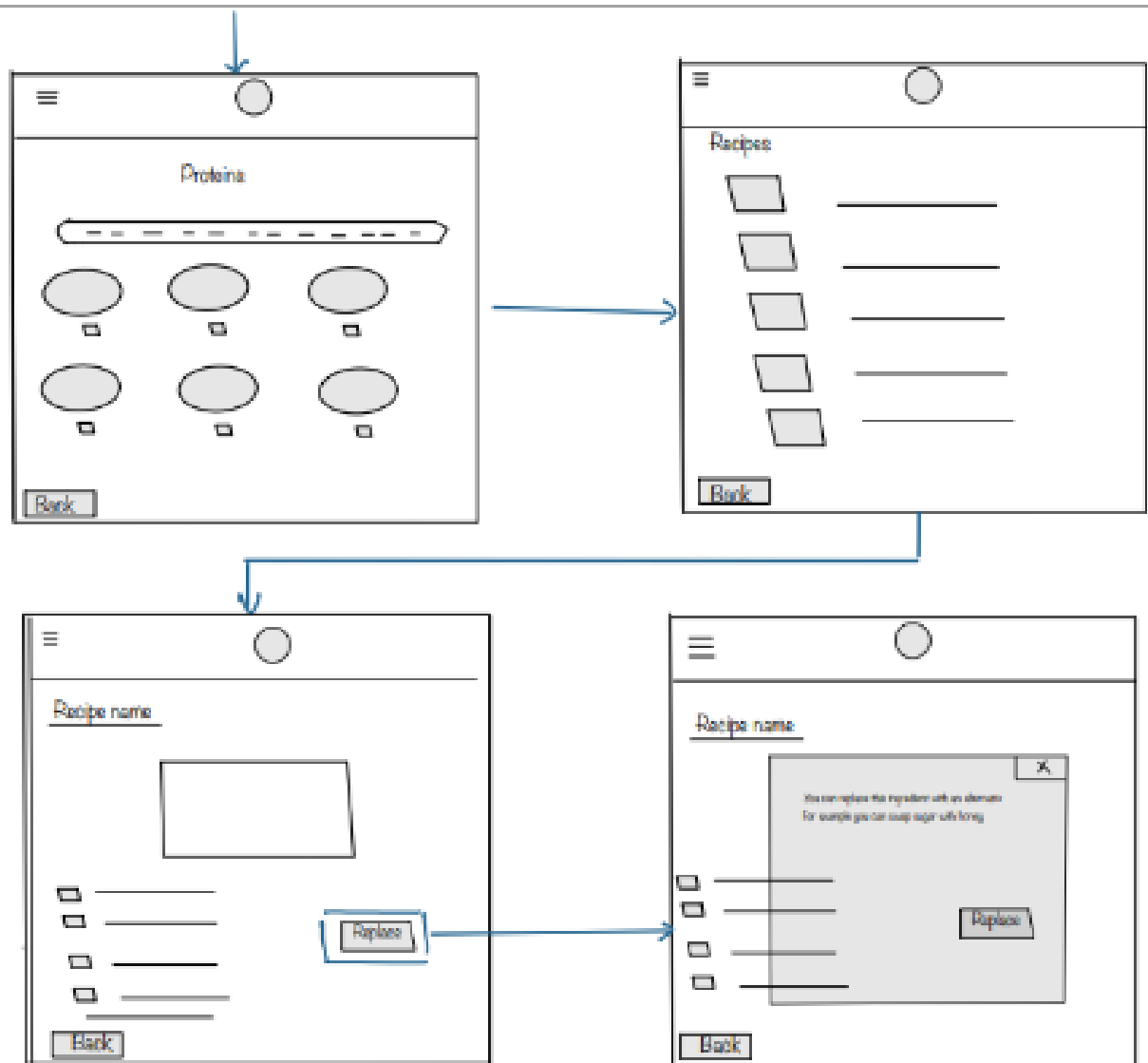
Task 3:

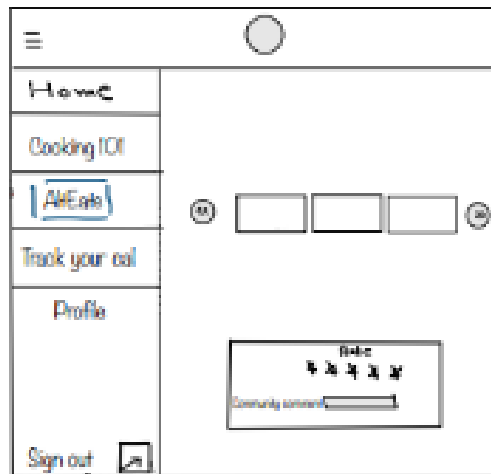
Develop an app that helps to track calories and nutritional information accurately, enabling h

14. The Low fidelity Prototype



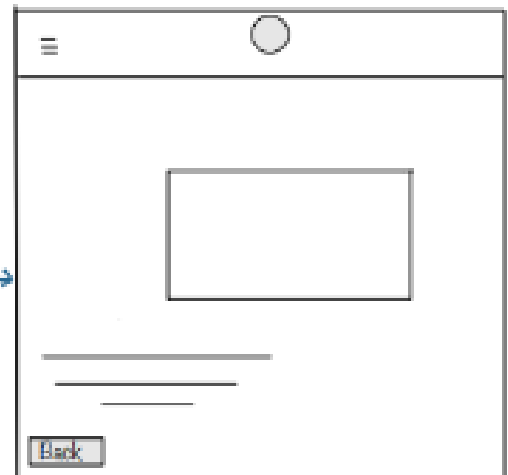
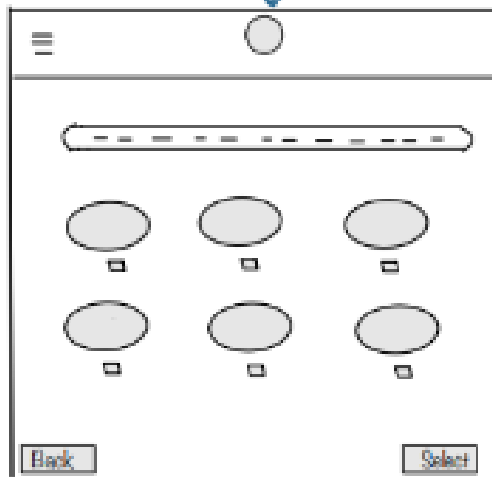


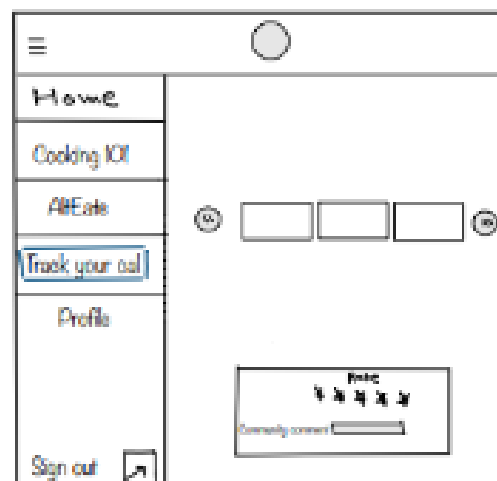




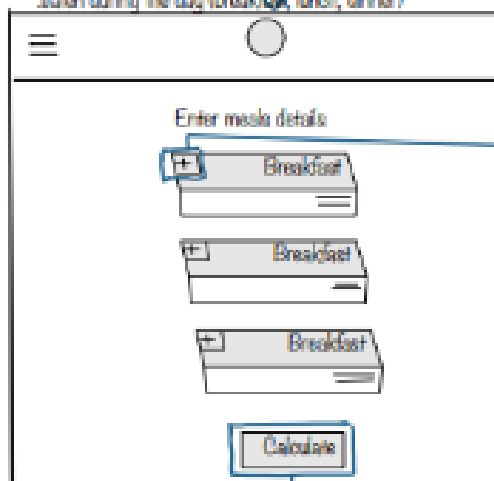
Task 2:

When the user selects AllEats, they can choose ingredients they are allergic to, and the app shows suitable alternatives

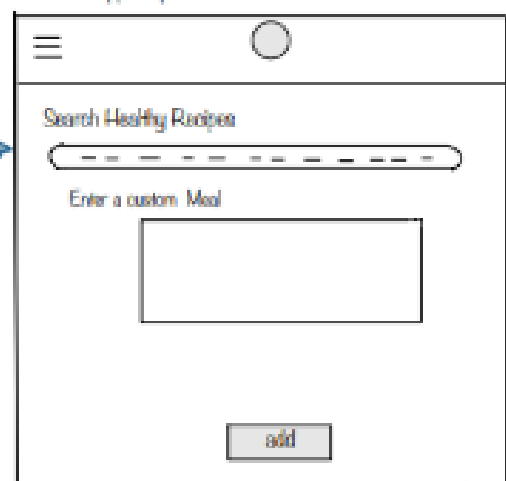




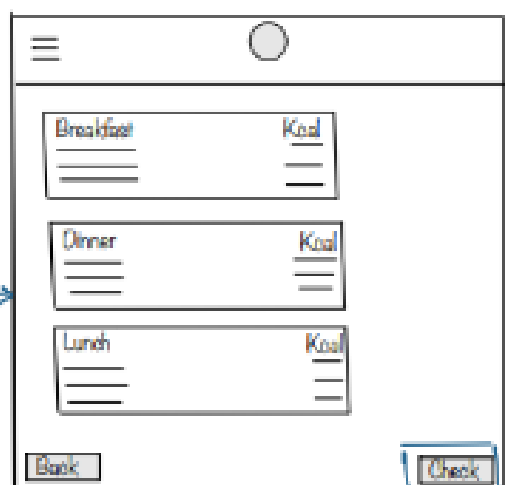
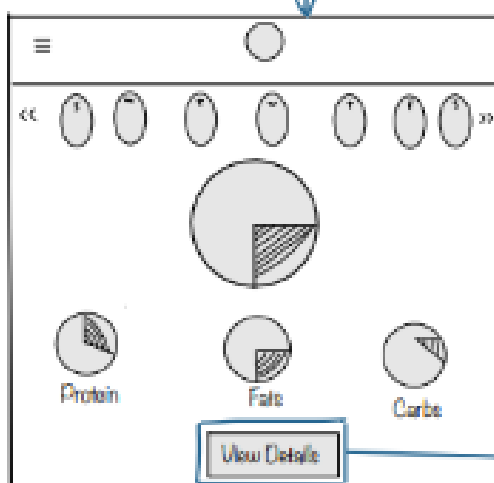
Task 3: When the user clicks TrackYourCal, a page opens to enter meals eaten during the day (breakfast, lunch, dinner)

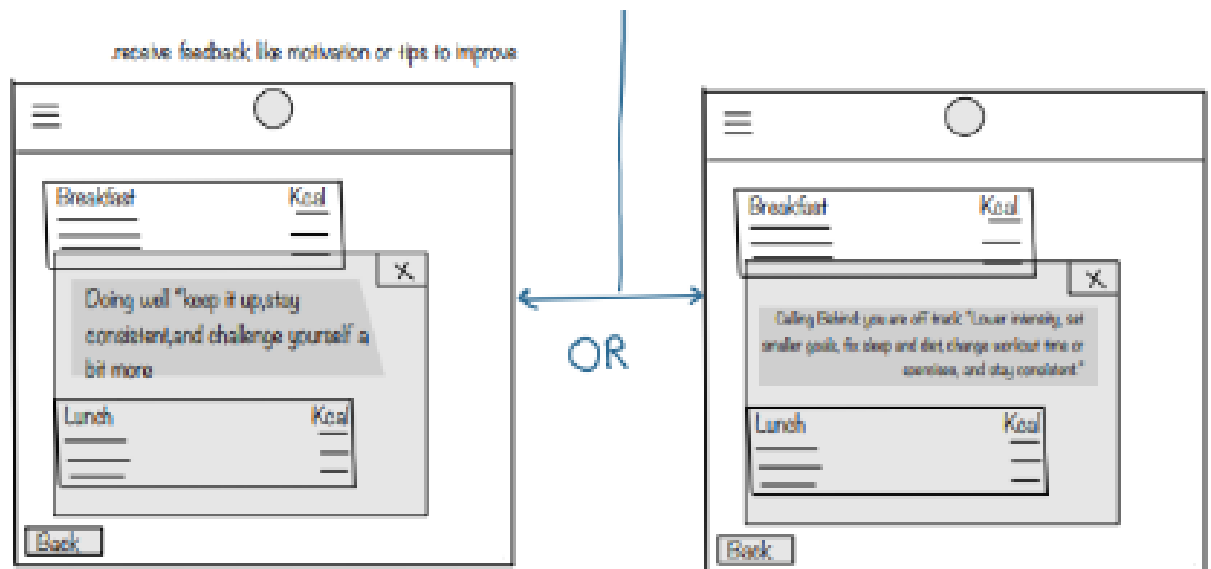


By clicking the plus sign, they can search app recipes or add their own



Tracks calorie intake and the nutritional breakdown of meals (protein, carbs, fats)





15. A paragraph explaining the new design and how it will meet users requirements

The design of The Fresh Bite App enhances user experience by making healthy eating simpler, smarter, and more accessible. Key features like Cooking 101, AltEats, and Track YourCal ensure that users- whether beginner cooks, health-conscious individuals, or fitness enthusiasts-can seamlessly integrate nutritious meals into their daily routines.

The Cooking 101 feature serves as a guided learning tool for beginners, providing step-by-step cooking instructions and tips, helping users build confidence in preparing homemade meals. The AltEats feature allows users to easily swap ingredients based on dietary restrictions, allergies, or availability, ensuring flexibility and inclusivity in meal

preparation. Meanwhile, TrackYourCal provides real-time calorie and macronutrient tracking, enabling users to monitor their nutritional intake and stay on top of their health goals.

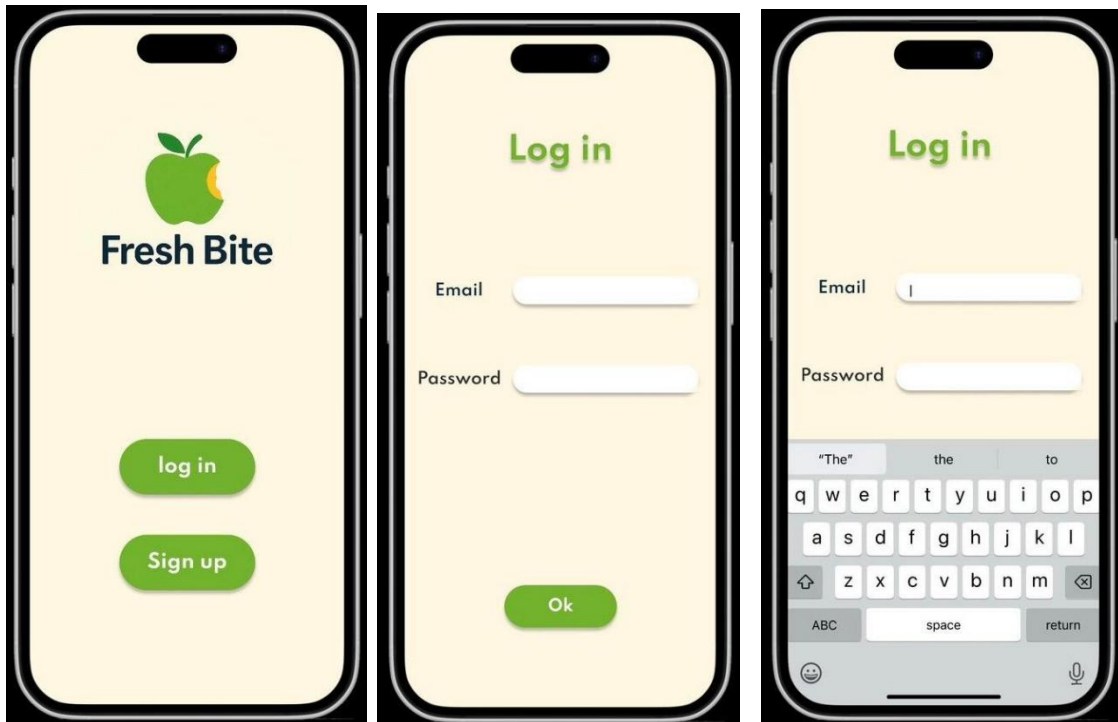
The app's improved navigation and interface allow users to input ingredients they have at home, receive personalized recipe suggestions, and categorize meals into breakfast, lunch, dinner, and snacks for better meal planning. A community comment section encourages user interaction, fostering a supportive space for sharing experiences and tips. With these enhancements, The Fresh Bite App makes healthy eating more convenient, educational, and tailored to individual needs.

16. The High fidelity Prototype

Log in :

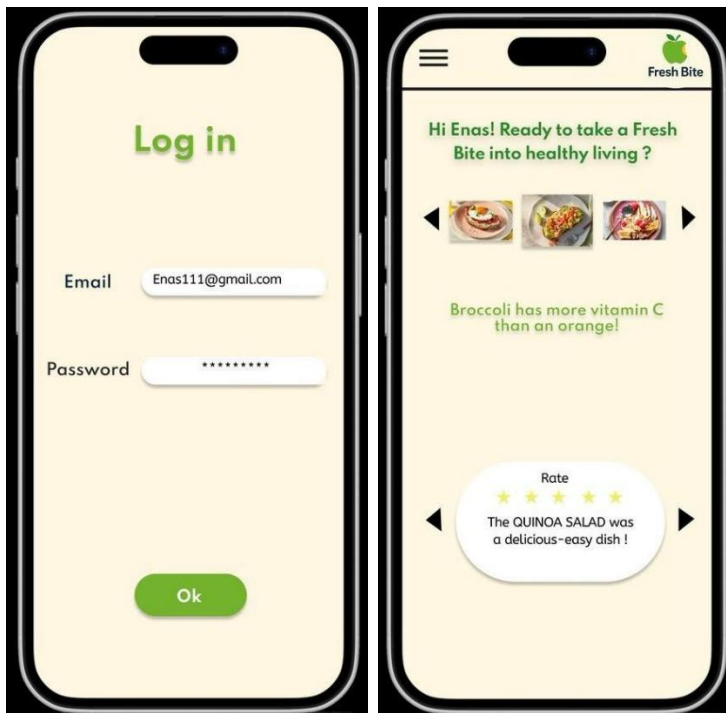
The user can click either log in or Sign Up

If the user chooses Log In



they need to enter their email and password

Home page



Sign up :

If the user chooses Sign Up

The first screen displays the 'Fresh Bite' logo and two buttons: 'log in' and 'Sign up'. The second screen shows the 'Sign up' form with the following fields: Email, Password, Gender (Female, Male), Age, Weight, Height, and Activity level (High, Medium, Low). The third screen shows the same form with a keyboard overlay.

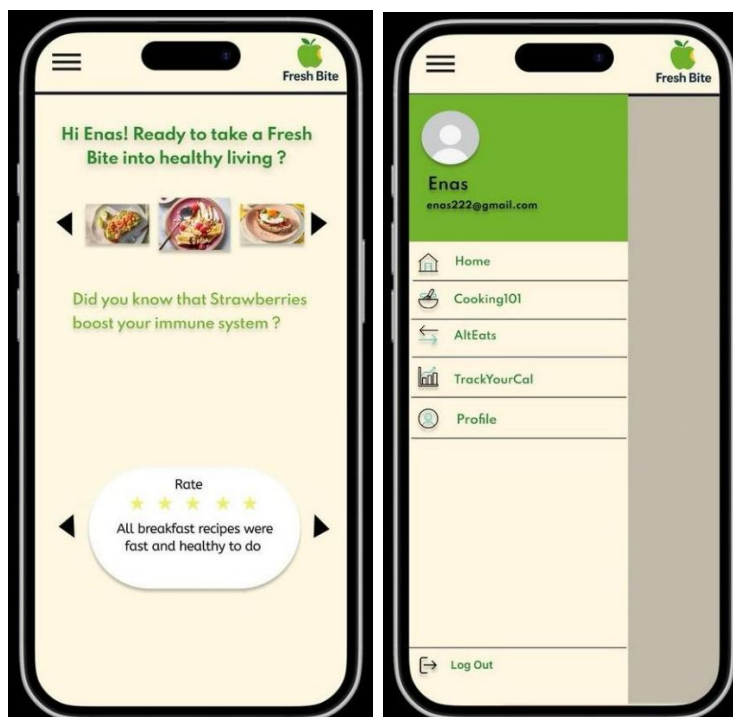
they must fill in the required information Home page

The first screen shows the 'Sign up' form with the following fields: Email (enas222@gmail.com), Password (*****), Gender (Female, Male), Age (20), Weight (47), Height (160), and Activity level (High, Medium, Low). The second screen shows the 'Home' page with a greeting 'Hi Enas! Ready to take a Fresh Bite into healthy living?', a carousel of food images, a quote 'Broccoli has more vitamin C than an orange!', and a rating section for 'The QUINOA SALAD' with a 5-star rating and the text 'The QUINOA SALAD was a delicious-easy dish!'.

Home page :

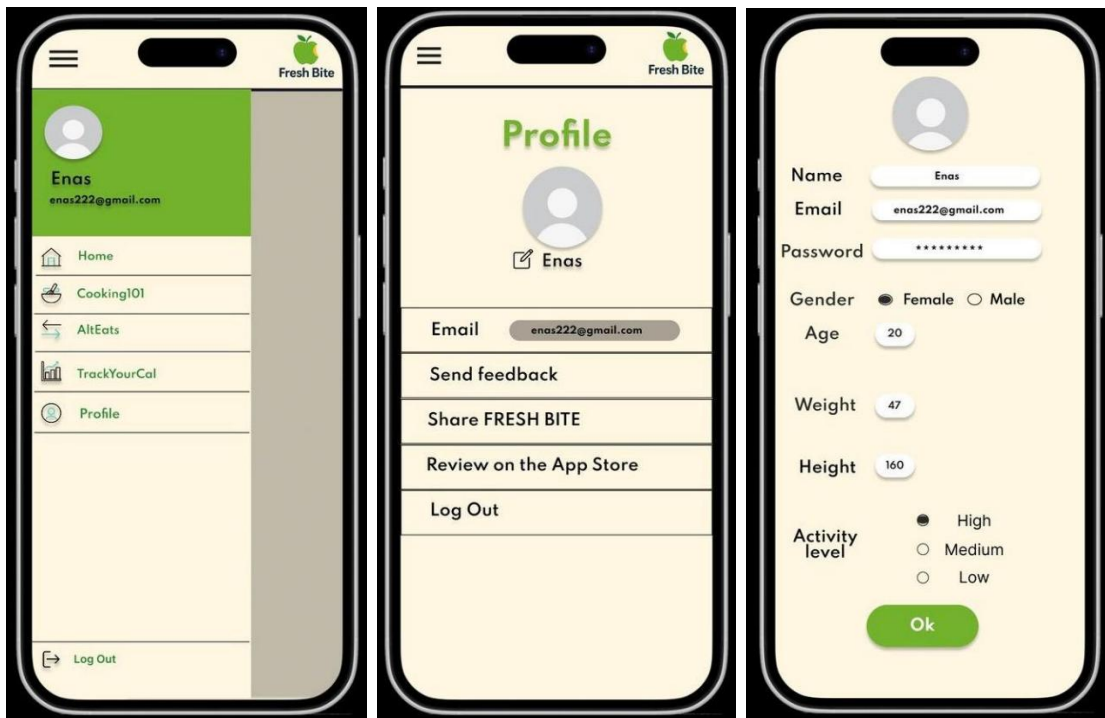


The Home Page displays different options like Recipes, Track Calories, and Profile



Profile page :

If the user clicks on Profile they will be directed to their personal page the user can click the “Edit” button to modify their selected

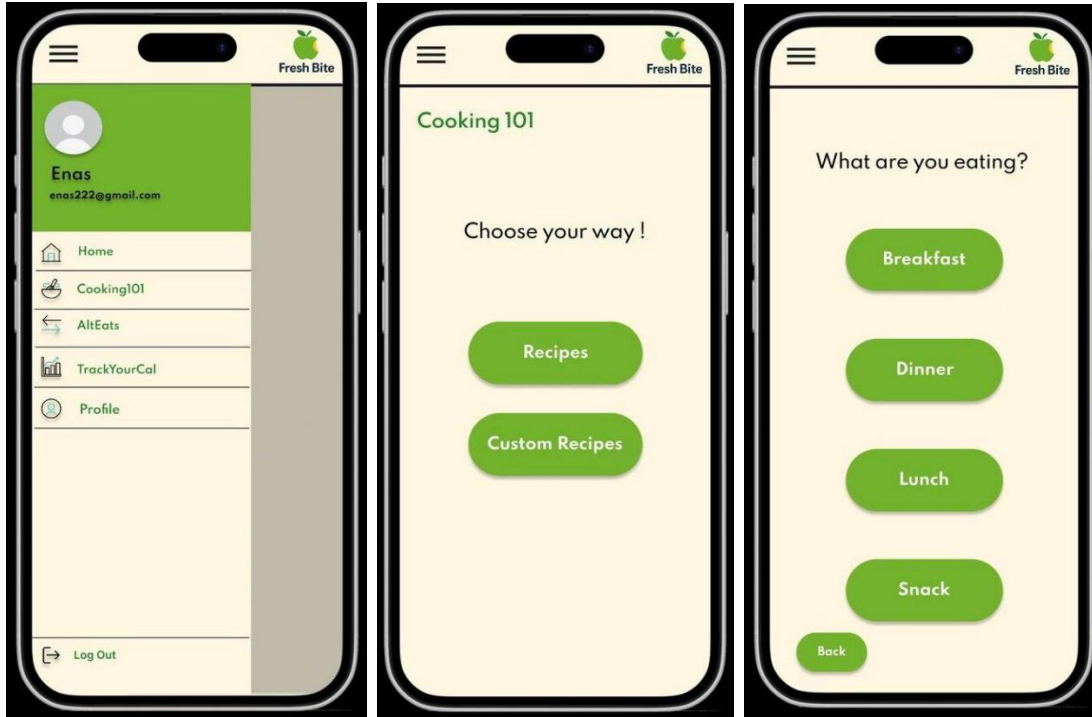


If the user sends feedback The user can submit their comment and rate the recipes



Task1 (cooking 101) :

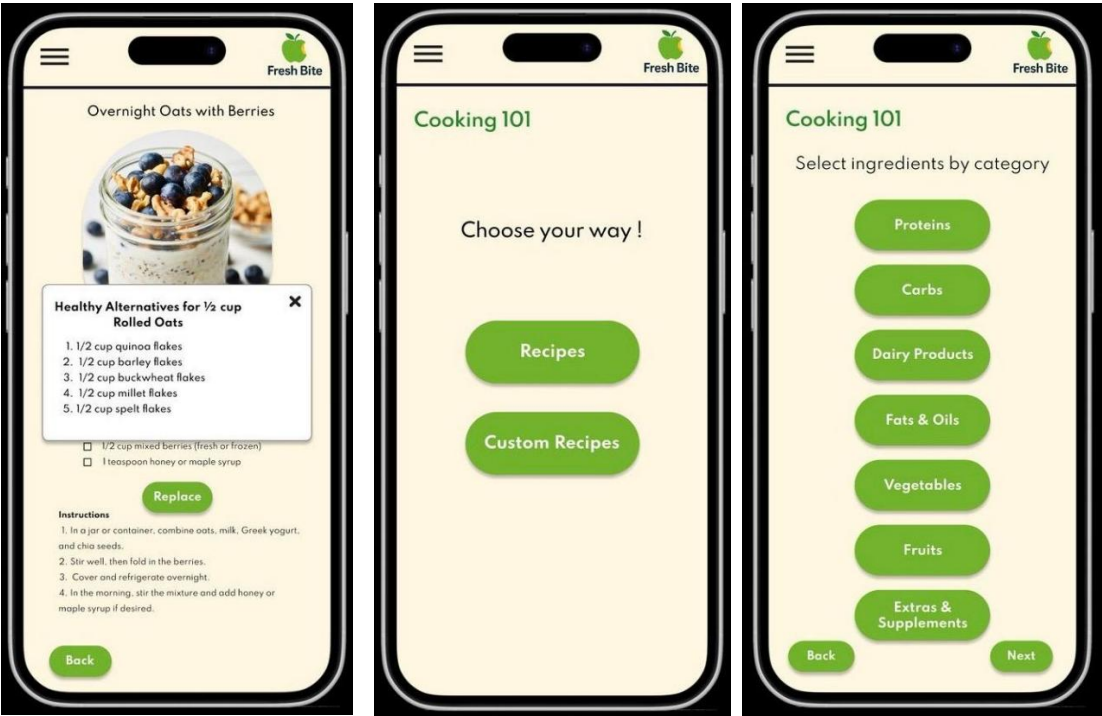
On the cooking 101 page, choose ready-made recipes, or customize their own recipes by selecting ingredient When the user presses Recipes shows meal time



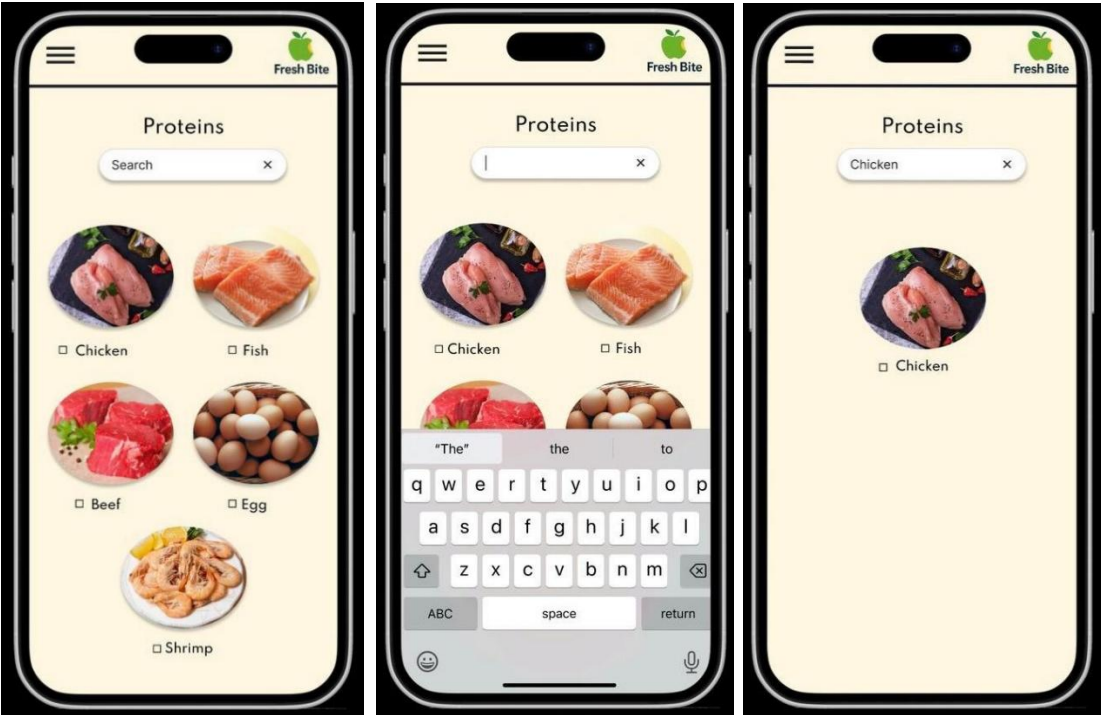
Shows matching recipes selecting a recipe edit ingredients by checkbox



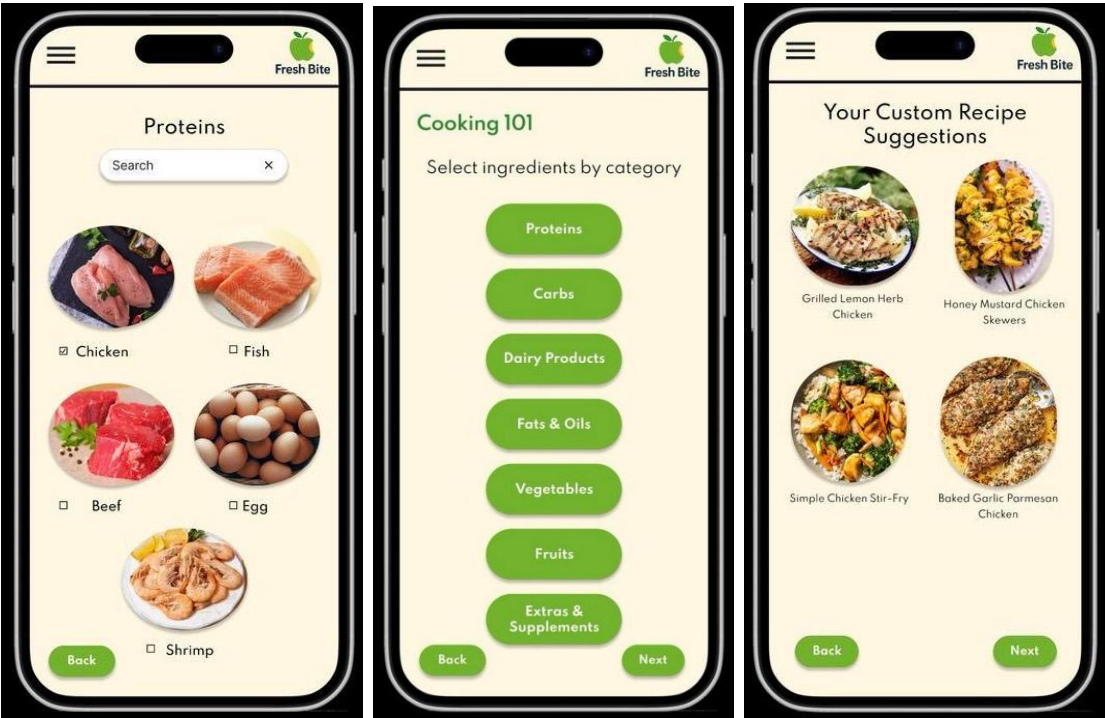
If the user selects custom recipes The user can enter the ingredients, they have available.



The user can press the ingredients through the checkbox or the user can search for the ingredients



After the user finishes selecting the desired ingredients click on “Next” button, shown suggested recipes that are closest to the ingredients.



The user chooses the recipe the user can edit the ingredients by clicking on the checkboxes. Then, by pressing the “Replace” button message will appear suggesting possible ingredient replacements.

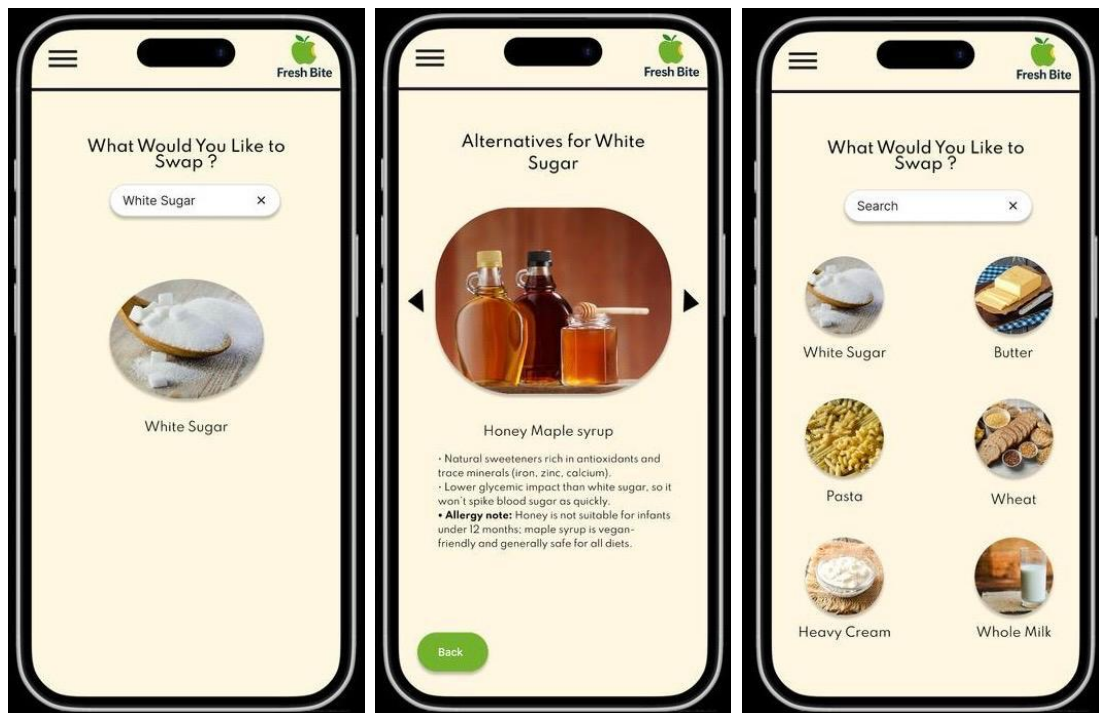


Task2 (Alt Eats) :

When the user selects AltEats they can choose ingredients or search



app shows suitable alternatives.

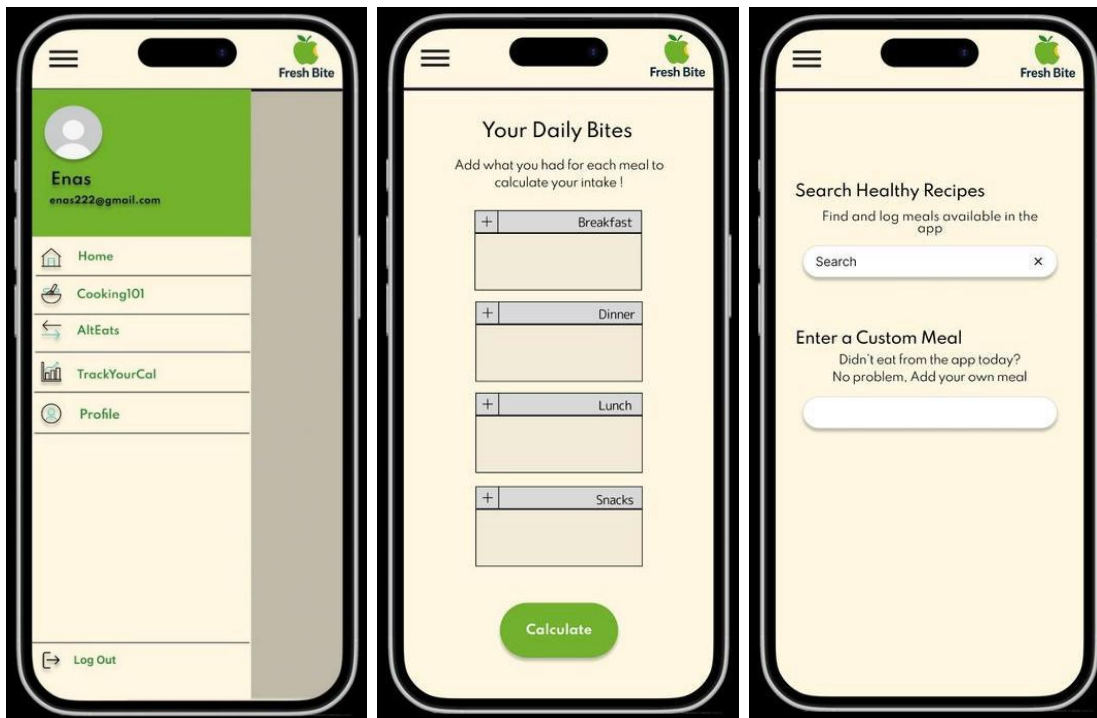


The user can browse multiple alternative options using the arrow buttons

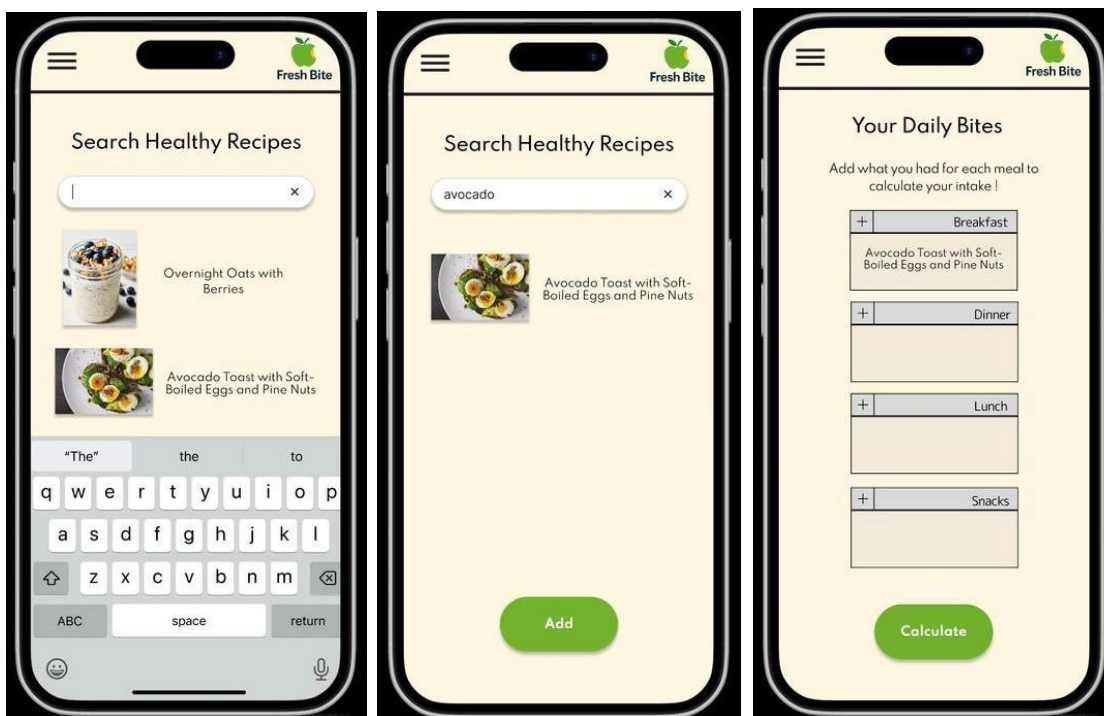


Task3 (Track Your Cal) :

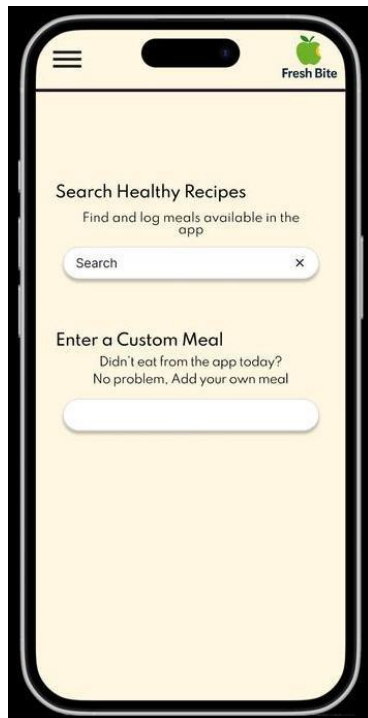
When the user clicks on Track Your Cal can enter their meal details. If they click the "+" button



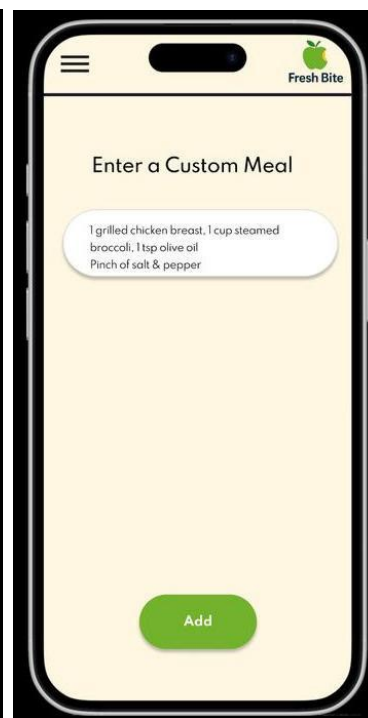
they can either search for recipes within the app, Add to Calculate the recipe details will appear under each meal.



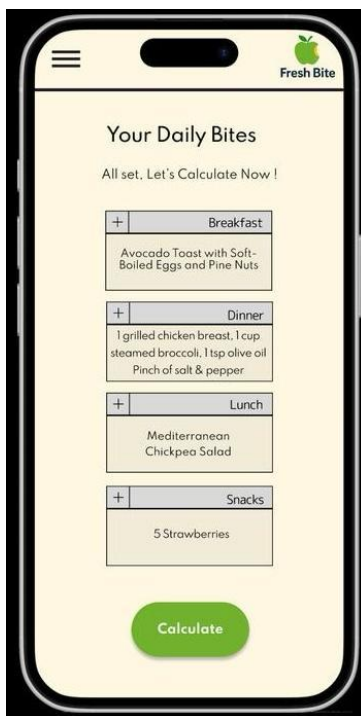
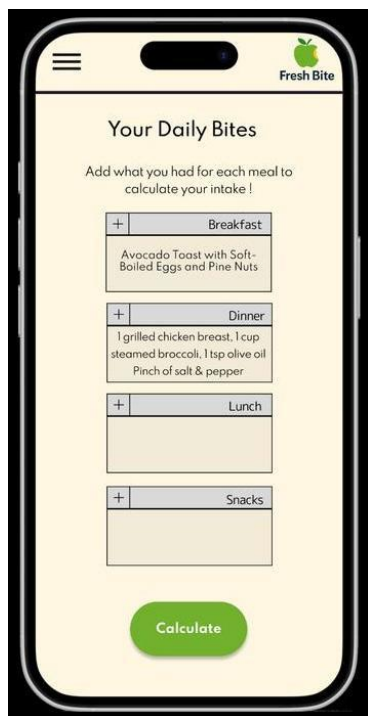
Add their own custom meals.



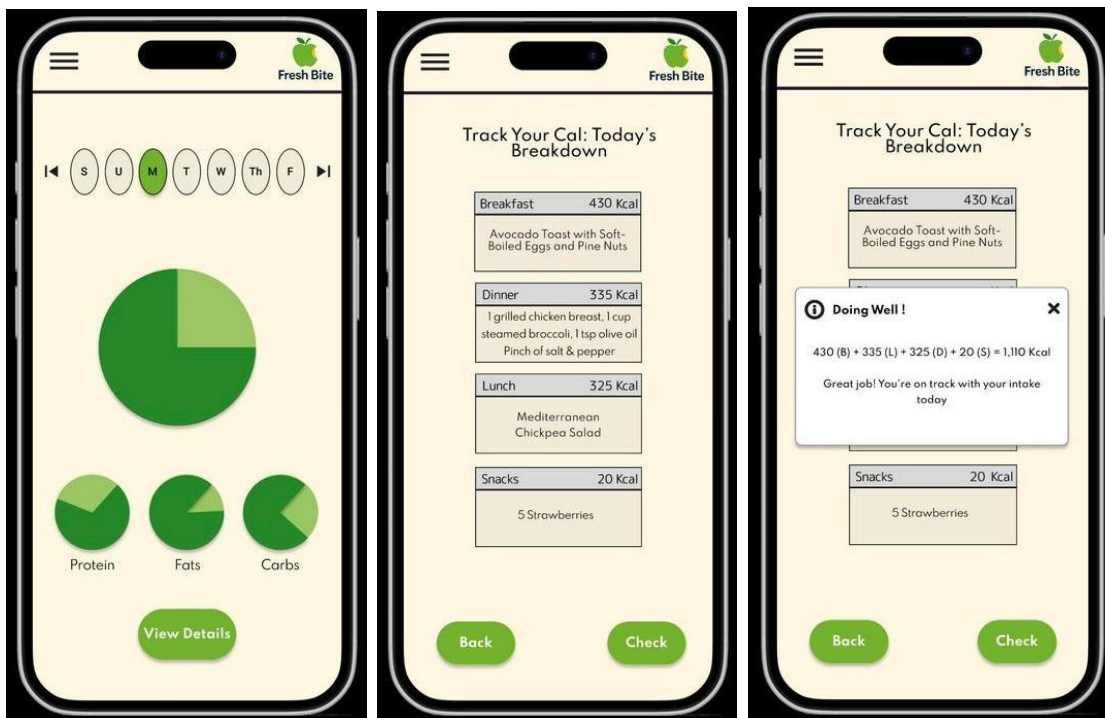
Add to Calculate



After completing the meal details, they can press the "Calculate" button



Show The total calories by clicking "View Details," they will be able to see a breakdown of calories for each meal



If the user selects "Check," a message receive feedback like motivation or tips to improve

